

DG4NLP Natural Language Processing

by Dr Amal Htait

Week 4: Coursework – Section 4

In this section, you will extend your previous NLP pipeline by creating an interactive interface for your system.

The aim is to transform your NLP and LLM experiments into a usable application that allows users to interact with your models more easily.

You may choose the complexity level of your interface depending on your interests and technical confidence. The interface can range from a simple graphical interface with buttons to a fully interactive chatbot system.

The focus of this section is not only functionality, but also usability, interaction design, and the integration of NLP components into an end-user application.

Task Overview

You will build an interactive NLP application that integrates your previous work. Your interface may include:

- A simple interface with buttons and text input/output areas
- A basic chatbot interface
- An advanced chatbot with memory, retrieval, or multi-step interaction

Steps

Step 1: Design the User Interface

You should design a simple and clear interface that allows users to interact with your NLP system.

Your interface may include:

- Text input fields
- Buttons to trigger tasks
- Areas to display predictions or generated outputs
- Dropdown menus or selectable options
- Chat-style interaction windows

Possible tools/frameworks include:

- Gradio

- Streamlit
- Flask
- FastAPI
- Any other suitable framework

You should explain:

- Why you selected your framework
- The design choices you made
- How users interact with your system

Step 2: Integrate NLP/LLM Components

You must connect your interface to the NLP models developed in previous coursework sections.

Possible functionalities include:

- Scientific paper classification
- Abstract summarisation/Title generation
- Conversational chatbot interaction

Your application should allow the user to submit input and receive model-generated outputs through the interface.

You are encouraged to experiment with different interaction styles and evaluate their usability.

Step 3: Testing and Evaluation

You should test your interface and evaluate:

- Functionality
- Ease of use
- Responsiveness
- Accuracy of outputs
- User experience

You may also compare:

- Different interface designs
- Different prompting styles
- Chatbot vs non-chatbot interaction

Coursework – Section 4 Outcome

Code

Clean and structured implementation of:

- User interface
- NLP/LLM integration
- Chatbot interaction (if applicable)
- RAG integration (if applicable)

Well-commented notebook/code

Documentation

Methods

A clear explanation of:

- Interface architecture
- Framework selection
- NLP/LLM integration
- Chatbot design (if applicable)
- RAG integration (if applicable)

Experiments and Evaluation

- Interface testing
- Interaction experiments
- Prompt comparisons
- User experience evaluation
- Comparison of different interaction methods